

Patternprinting

```
In [3]:
data
=
"Python Programming"
print
(
data
[
0
])
print
(
data
[
-
1
])
print
(
data
[
0
:
6
])
data
[
0
]
=
"A"
P
g
Python
-----
TypeError
Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_1972\1187066189.py
in
<module>
3
print
(
data
[
-
1
]
)
4
print
(
data
[
0
:
6
]
)
```

```
----> 5
data
[
0
]
=
"A"
TypeError
: 'str' object does not support item assignment
```

```
In [4]:
data
=
"Python Programming"
print
(
data
*
3
)
Python ProgrammingPython ProgrammingPython Programming
```

```
In [5]:
data
=
"Python Programming"
print
(
data
in
'U'
)
print
(
data
not
in
'U'
)
False
True
```

```
In [6]:
data
=
"Python Programming"
print
(
data
.
upper
())
PYTHON PROGRAMMING
```

```
In [8]:
data
=
"Python Programming"
print
(
data
.

```

```
lower
()  
python programming
```

```
In [ ]:  
data  
=  
"Python Programming"
```

```
In [13]:
```

```
row  
=  
4  
column  
=  
4  
for  
i  
in  
range  
(  
1  
,  
row  
+  
1  
):  
for  
j  
in  
range  
(  
1  
,  
column  
+  
1  
):  
print  
(  
"*"  
,  
end  
=  
" "  
)  
print  
(  
end  
=  
"  
\n"  
)  
* * * *  
* * * *  
* * * *  
* * * *
```

```
In [14]:
```

```
row  
=
```

```

4
column
=
4
star
=
"*"
for
i
in
range
(
1
,
row
+
1
):
for
j
in
range
(
1
,
column
+
1
):
print
(
star
,
end
=
" "
)
print
(
end
=
"\n"
)
* * * *
* * * *
* * * *
* * * *

```

```

In [15]:
n
=
4
star
=
"*"
for
i
in
range

```

```

(
1
,
n
+
1
):
for
j
in
range
(
1
,
n
+
1
):
print
(
star
,
end
=
" "
)
print
(
end
=
"
\n
"
)
* * * *
* * * *
* * * *
* * * *

```

```

In [16]:
n
=
6
star
=
"*"
for
i
in
range
(
1
,
n
+
1
):
for
j
in
range

```

```
(
1
,
n
+
1
):
print
(
star
,
end
=
" "
)
print
(
end
=
"
\n
"
)
*****
*****
*****
*****
*****
*****
```

```
In [17]:
m
=
4
n
=
6
star
=
"*"
for
i
in
range
(
1
,
m
+
1
):
for
j
in
range
(
1
,
n
+
1
```

```
):
print
(
star
,
end
=
" "
)
print
(
end
=
"
\n
"
)
*****
*****
*****
*****
```

```
In [24]:
n
=
5
for
i
in
range
(
n
,
0
,
-
1
):
for
j
in
range
(
i
):
print
(
"*"
,
end
=
" "
)
print
(
end
=
"
\n
"
)
```

```
*****
*****
***
**
*
```

In [26]:

```
n
=
5
for
i
in
range
(
1
,
n
+
1
,
1
):
for
j
in
range
(
i
):
print
(
"*"
,
end
=
" "
)
print
(
end
=
"\n"
)
*
*
*
*
*
*
*
*
*
*
```

In [32]:

```
n
=
5
for
i
in
range
(
```

```
1
,
n
+
1
,
1
):
for
j
in
range
(
1
,
i
+
1
):
print
(
j
,
end
=
" "
)
print
(
end
=
"
\n
"
)
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

In []: